LESSON 1.5

OPERATIONS WITH FRACTIONS

LESSON INTRODUCTION

MA.5.AR.1.2 Solve real-world problems involving the addition, subtraction, or multiplication of fractions, including mixed numbers and fractions greater than one.

MA.5.AR.1.3 Solve real-world problems involving division of a unit fraction by a whole number and a whole number by a unit fraction.

LEARNING TARGETS

- I can solve real-world problems involving adding and subtracting fractions.
- I can solve real-world problems involving multiplication of fractions.
- I can solve real-word problems involving division of a unit fraction by a whole number.
- I can solve real-world problems involving division of a whole number by a unit fraction.

BENCHMARK COHERENCE

WORKING TOWARD

MA.6.NSO.2.3 Solve multistep real-world problems involving any of the four operations with positive multidigit decimals or positive fractions, including mixed numbers.

ESSENTIAL QUESTIONS

- How can the information in a word problem to write an expression that can be used to determine the answer?
- What strategies can be used to determine what operations are used to solve a real-world problem?

LESSON NARRATIVE

In this lesson, students solve real-world problems involving operations with fractions. Students practice identifying which operation to use and what steps are required to solve, using any strategy including models and algorithms to solve. The Guided Practice problems involve all operations, which position students to practice skills from the previous lessons related to writing accurate expressions and solving when dividing and multiplying fractions. The Wrap-Up gives students an opportunity to reflect on learning and how a Growth Mindset relates to challenges in math.

COMPONENT SUMMARY

1.5.1 WARM-UP *(~5 MIN.)*

Critical thinking task on creating fraction multiplication with a product that is a mixed number

1.5.2 GUIDED INSTRUCTION *(15-20 MIN.)*

Guided instruction on choosing correct operation and steps to solve fraction problems in real-world context

1.5.3 GUIDED PRACTICE *(15-20 MIN.)*

Students solve real-world one- and two-step problems related to a real-world context

1.5.4 WRAP-UP *(~5 MIN.)*

Growth Mindset reflection on a math challenge and how to persevere in the future

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Multistep problem

MATERIALS

- N/A

ADDITIONAL PREPARATION

- Before the lesson, consider researching or having a list of popular book and movie characters for students to reference during the Growth Mindset reflection in 1.5.4 Wrap-Up.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have difficulty understanding which operation to use when solving a word problem.
- Students may look for key words to indicate which operation to use.
- Students may incorrectly represent a real-world situation with the divisor/dividend switched in division problems.

THINGS TO CONSIDER

- Be flexible with the use of the calculator. This lesson is focused on whether students can use the context of the situation to identify what operation is most appropriate and efficient to use.
- Operations with fractions skills are revisited through the spiral learning embedded in the scope and sequence. During this lesson, focus student learning on identifying which operation to use and using appropriate models or methods to solve.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

1.5.2 Guided Instruction positions students to identify which operation to use to solve one- and two-step problems. Consider using Think-Pair-Share literacy routines as students make decisions on which operation to use. This collaborative approach allows students an opportunity to attempt problems individually and receive support from a partner to confirm answers and hear a different perspective on how to approach solving real-world fraction problems.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

1.5.3 Guided Practice provides multiple opportunities for students to select efficient and appropriate methods for solving fraction operation problems within the given context. After ensuring that students understand the context provided, facilitate discussion that connects the relationships in the context to the operations and steps needed to solve the problem.

DIFFERENTIATE INSTRUCTION

During 1.5.2 Guided Instruction, support understanding the context of the word problems. Consider building background knowledge and building vocabulary by showing pictures where possible of the different scenarios. Preview the word problems for any specific vocabulary that may hinder comprehension by students. Consider removing the number values; students are reading the problems for context only.

Consider having students create an organized list of steps needed to complete multistep problems throughout the lesson. This visual support may also help students identify the steps and operations needed in their arithmetic. In addition, consider encouraging visual models to provide clarity for students who have difficulty with reading comprehension in story problems.

Though not included in the answers in this guide, consider allowing students to create models or pictorial representations of the problem context to help them organize their thinking and show their answers during 1.5.3 Guided Practice.

1.5.1 WARM-UP: CRITICAL THINKING

Component Overview: *Have students read the prompt and complete the task individually for about five minutes, then bring the class together to share and discuss a few options that are solutions. Focus the discussion on reviewing how to multiply and divide fractions, as the skill will be needed in coming sections of the lesson.*

ENRICHMENT

For students who may have difficulty getting started, ask questions to promote fraction reasoning and thinking.

- What do you know about multiplying fractions?
- What do you know about dividing fractions?
- How could one of the fractions be changed so that the solutions will be the same?

1.5.1 Warm-Up: Critical Thinking

1. **Place a digit in each blue box to make the solutions to the two problems the same. Use only digits 1 to 9, at most one time each.**
Answers will vary.

$$\frac{\boxed{9}}{\boxed{4}} \times \frac{\boxed{7}}{\boxed{3}} = \boxed{5}\,\frac{\boxed{2}}{\boxed{8}}$$

$$\frac{\boxed{9}}{\boxed{4}} \div \frac{\boxed{3}}{\boxed{7}} = \boxed{5}\,\frac{\boxed{2}}{\boxed{8}}$$

1.5.2 Guided Instruction: Operations with Fractions

The Gomez family decided to take a family vacation to the Walt Disney Resort in Florida. They had a jam-packed week planned with a lot of fun activities.

1. The Gomez family first went to Typhoon Lagoon to enjoy the waterpark. When they arrived, the air temperature in degrees Fahrenheit (°F), was $84\frac{1}{2}$°F. Every hour that they stayed, the air temperature increased $1\frac{1}{2}$°F. The family stayed $4\frac{3}{4}$ hours.

   a. Select a question that could be answered using the information given.

   - o What time did the family arrive?
   - o What was the temperature when they left?

To answer the question, more than one operation will need to be applied. When a word problem uses more than one step it is known as a **multi-step** problem.

   b. The first step could be to [o add / o multiply / o subtract / o divide] the number of hours the family spent at the waterpark by the value [o $1\frac{1}{2}$. / o $84\frac{1}{2}$.]

   c. Write the expression that represents the first step.

   $4\frac{3}{4} \cdot 1\frac{1}{2}$

1.5.2 GUIDED INSTRUCTION: OPERATIONS WITH FRACTIONS

Component Overview: *Students are guided through a real-world scenario that involves operations with fractions.*

TEACHER MOVES

Consider solving question 1 as a class, then use Think-Pair-Share routines as students work in partners to complete question 2, with an opportunity to regroup as a class at the end of each question to discuss solutions or remaining questions. Students can solve question 3 individually or in partners depending on student needs.

DIFFERENTIATED INSTRUCTION

If students have trouble deciding which question is more appropriate, prompt students to identify the kind of information the problem provides and match what question they can answer using that information.

QUESTIONING

For 1b, challenge students to explain why multiplication is the best operation to answer this question as opposed to addition.

1.5.2 GUIDED INSTRUCTION: OPERATIONS WITH FRACTIONS (CONTINUED)

ENRICHMENT

To develop awareness of multistep problems and the thinking required to identifying the steps, ask questions like:

- What does each value represent in the context of the story?
- How does re-reading the question help you determine if you have finished all the steps needed to solve?

TEACHER MOVES

Think-Pair-Share

Consider using Think-Pair-Share routines for all parts of questions 2a-2c. Give students time to attempt the questions individually, then review their responses with a partner. Have partners work together to fill in the table on question 2d on what they noticed between the multistep problem in question 1 and the one-step problem in question 2. Consider also providing an opportunity for partnerships to share their findings with the class, as correctly identifying operations and steps in real-world problems is the focus of this lesson.

ENRICHMENT

For question 2c, prompt partners to consider:

- How does your expression represent the context related to wait time?

QUESTIONING

Early finishers can create a model to represent the division question in 3a. Consider having students share their models as a review of previous lessons and to justify their solution using the algorithm.

d. The next step is to use the value that is the result of the first step and [o add it to / o multiply it with / o subtract it from / o divide it into] the value [o $1\frac{1}{2}$. / o $84\frac{1}{2}$]

e. Write the expression that represents the first step. Then, perform the operation and write the value that is the result of the second step.
$4\frac{3}{4} \cdot 1\frac{1}{2}; 91\frac{5}{8}$

f. Reread the problem. Explain why the result of the second step makes sense for the question you chose.
Answers will vary. The answer makes sense because it shows that the temperature increased. The answer makes sense because the temperature increased by over 5°.

2. While at the waterpark, Mr. Gomez kept track of the time the family spent waiting in line for rides. At the Humunga Kowabunga slide, they waited $\frac{5}{6}$ of an hour. At Crush 'n' Gusher, they waited $1\frac{1}{4}$ hours in line. How much longer did they wait at Crush 'n' Gusher than at Humunga Kowabunga?

a. Which of the following correctly restates the information the question is asking for?

   o Find the total time the family waited in line for Humunga Kowabunga and Crush 'n' Gusher.
   o Find the difference in the family's wait times for Humunga Kowabunga and Crush 'n' Gusher.

b. To answer the question, [o add / o multiply / o subtract / o divide] the family's wait times for Humunga Kowabunga and Crush 'n' Gusher.

c. Write the expression that represents the context. Then, perform the operation and write the value that is the result.
$1\frac{1}{4} - \frac{5}{6}; \frac{5}{12}$

d. Discuss with your partner one thing you noticed between the multi-step problem and the problem that used just one step. Write your observations in the table.

I noticed...	My partner noticed...
Answers will vary. I noticed that the multi-step problem had 3 numbers. I noticed that the one-step problem had 2 numbers.	*Answers will vary.*

3. The three Gomez children wanted a snack. Their mom gave them a bag of chips that was approximately $\frac{1}{2}$ full. The children divided the half-full bag of chips equally among the 3 of them.

a. Work with your partner to write a question that makes sense for the context and uses the values $\frac{1}{2}$ and 3.
Answers will vary. How much of the bag of chips did each child get?

b. Write the answer to your question and show the work necessary to get the answer.
$\frac{1}{6}; \frac{1}{2} \div 3 = \frac{1}{6}$

1.5.3 Guided Practice: Operations with Fractions

The Gomez family spent their second day of vacation at the Magic Kingdom. Vario Gomez and his dad left the rest of the family to ride some of the more adventurous rides. A list of the rides in the order they rode them and the distance they walked to get to each ride is shown below.

Order	Ride	Distance Walked (in miles)
1	Splash Mountain	$\frac{3}{4}$
2	Tom Sawyer's Island	$\frac{1}{6}$
3	Seven Dwarfs Mine Train	$\frac{1}{2}$
4	Big Thunder Mountain Railroad	$\frac{3}{4}$
5	Space Mountain	$1\frac{1}{12}$

Use the table to answer each of the following questions. Write the expression or equation you use.

1. Vario tracks his steps using a fitness tracker. He averages 2,000 steps for each mile. How many total steps would Vario and his dad take as they walked to the rides listed in the table?
   $(3/4 + 1/6 + 1/2 + 3/4 + 13/12) \cdot 2{,}000 = 6{,}500$ *steps*
2. How much farther, in miles, was the walk to Space Mountain than to Splash Mountain?
   $\frac{13}{12} - \frac{3}{4} = \frac{1}{3}$ *of a mile*

1.5.3 GUIDED PRACTICE: OPERATIONS WITH FRACTIONS

Component Overview: *Working in partners or individually, students create and solve expressions related to the same context from the Guided Instruction section.*

TEACHER MOVES

After each question, prompt students to share why they chose the operation and the equation, expression, or representation they created to answer the question.

DIFFERENTIATED INSTRUCTION

Go Numberless

If students are having difficulty identifying what operation to use, consider making the word problem numberless. Remove all the numerical values and read the word problem to discuss what is happening. With the absence of numbers, students can focus on the context, helping them to determine the operation.

1.5.3 GUIDED PRACTICE: OPERATIONS WITH FRACTIONS (CONTINUED)

TEACHER MOVES

Notice that question 5 is a multistep problem whereas questions 1-4 are one-step problems. Consider highlighting this distinction through discussion, asking questions like:

- What do you notice that is different for question 5?
- For question 5a, what part of the question's context tells you that you need to solve using two steps?
- What operations and steps are needed to solve question 5b? How does 5b use some of the steps/solutions you found in 5a?

ENRICHMENT

For question 5a, support reasoning about multistep problems by asking:

- How can you first determine how much time they spent swimming? What operation could be used?
- After you know how much time was spent swimming, what do you need to find next? What operation could be used to find that value?

3. Mrs. Gomez and her two daughters waited in line a total of 2 hours to meet different characters. Each character's line was approximately $\frac{1}{5}$ of an hour long. How many characters did they get to meet?
$2 \div \frac{1}{5} = 10$ *characters*

4. Later in the day, the family met in Adventureland to go on some rides together as a family. They rode a total of 5 rides and spent $\frac{1}{2}$ hour at each ride. How many hours did they spend in Adventureland?
$5 \cdot \frac{1}{2} = 2\frac{1}{2}$ *hours*

5. On their third day of vacation, the Gomez family decided to stay at the resort. The family did the following activities.
- **Rented a bike for $2\frac{1}{4}$ hours.**
- **Swam in the pool 2 different times. Each time they swam for $\frac{5}{6}$ hours.**
- **Played on the playground 3 times. Each time was for $\frac{5}{12}$ hours.**

a. How many more hours did they spend riding bikes than swimming?
$\frac{7}{12}$ *of an hour*

b. How many hours did the family spend doing all of the activities?
$5\frac{1}{6}$ *hours*

1.5.4 Wrap-Up: Growth Mindset

1. Today, I was challenged by *Answers will vary.*

   When I am challenged I (o avoid / o embrace) the challenge.

   Today, I (o gave up easily. / o persisted in learning.)

2. Think about the characters in your favorite movie or book. Describe a time the character had to persist in the face of a challenge.
   Answers will vary.

3. Describe something that your teacher can do to help you overcome challenges you may face in math this year.
   Answers will vary.

4. Describe something that you can do to overcome challenges you may face in math this year.
   Answers will vary.

1.5.4 WRAP-UP: GROWTH MINDSET

Component Overview: *This Wrap-Up provides an opportunity for students to reflect about a challenge they encountered during this lesson, relate that challenge to a book or movie character's obstacles, and brainstorm an idea of how they will address that challenge in the future. This reflection also provides information on students' perceived ideas of what may be difficult for them this school year related to math and how the teacher may help.*

TEACHER MOVES

Students could identify a portion of today's math lesson during which they felt a challenge to help them with responding to questions 3 and 4.

Use the information that students give in the Wrap-Up to get to know your students more. Note each student's answer for question 3 to know what ways you can help them overcome challenges they may face during the year.

DIFFERENTIATED INSTRUCTION

Consider an alternate form of students communicating their responses to questions 3 and 4, like drawing a picture or comic with captions/text bubbles or sharing with the teacher verbally.